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Combinations

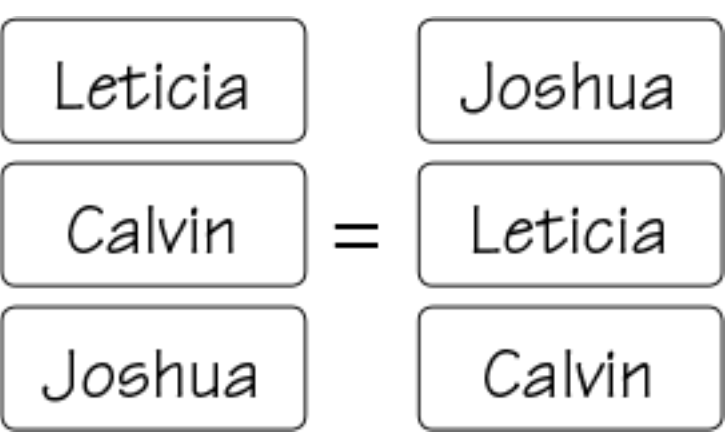
Toolbox

Combinations

A **combination** is an arrangement of the items in a set in which the order of the items doesn't matter.

Suppose there are 8 workers in an office. The fire safety team needs 3 people. The manager puts the 8 workers' names on slips of paper and randomly chooses 3 names. Does the order the names are chosen make a difference?

No, the workers will be a group. The order doesn't matter.



The orders are different, but the group of names is the same.

This is an example of a combination. Now let's see how to count it.

Example

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Three employees out of a group of 8 will be chosen randomly to be the fire safety team. How many different groupings are possible?

Step 1

Find the permutations for choosing 3 names out of 8.

$8 \times 7 \times 6 = 336$

Step 2

Find the permutations for a group of 3. You are finding how many different ways 3 names can be arranged.

$3 \times 2 \times 1 = 6 \rightarrow 3! = 3 \times 2 \times 1 = 6 \text{ ways}$

Step 3

Divide.

$336 \div 6 = 56$

There are 56 possible combinations.

A B C

A C B

C A B

B A C

B C A

C B A

How can you tell the difference between a permutation and a combination problem? You need to use common sense. Remember, if changing the order matters in the problem, it's a permutation.

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Lesson 4: Counting Problems

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